

Prediction of Prices in Agriculture (PREPARE)

Date: 22 March 2026

Project Overview

The main idea behind our project is to solve farmer's price blindness. Right now, small and marginal farmers don't have accurate, timely price forecasts to understand market dynamics. We also want to provide reliable multi-day price prediction tools so that the government and regulators can plan better market interventions.

Please refer to the previous report for semester 5 progress and progress made in semester 6 prior to the current report.

Progress Since the Last Report

Over the past two weeks, we focused on finalizing our datasets and establishing a solid baseline model. We used the Copernicus API to add 24-hour weather features for each day and mandi and we utilized Geopy to pull the exact location data for each mandi. With these additions, our final datasets are fully assembled.

A major focus during this sprint was addressing data sparsity. Instead of relying on a single guessing method, we researched and engineered ten distinct imputation strategies, ranging from simple baselines to advanced Matrix Factorization, to see which would work best for our pipeline.

The 10 Imputation Strategies Tested:

1. Capped Forward Fill: Carries the last known price forward. This is plausible as agricultural prices often remain stable for short periods. We capped it at a few days to prevent a stale price from polluting massive data gaps.
2. Rolling Mean: Fills gaps using the average of surrounding days within a 7-day or 15-day window. This effectively smooths out the daily volatility caused by localized reporting noise.
3. State Median: Imputes missing prices using the median price of all other active mandis in the same State on that exact day. This acts as a spatial proxy, assuming mandis within the same state face identical macro-conditions like weather and transport costs.
4. Simple Historical Seasonality: Uses historical calendar norms, grouping by month and day of the week, to fill gaps. This is logical because crops follow predictable planting and harvest cycles.
5. Global Crop Median: Calculates the overall median price for the crop across the entire 3-year dataset. This functions purely as a structural safety net to guarantee 100% data coverage if all other methods fail.
6. Forward Fill with Spatial Decay: Blends temporal proximity with spatial reality. As a missing gap gets longer, the weight of the last known price decays, while the weight of the State-level daily average increases.

7. DOW Ratio (Day of Week): Calculates the ratio of Sunday prices to the weekly average for a specific mandi. It imputes missing Sundays by applying that historical ratio to the surrounding week's average. Our EDA highlighted extreme structural missingness on specific days, making this highly relevant.
8. The Spline Pipeline: A 4-step approach that flags outliers, fits a year-wise cubic spline, runs window-based smoothing, and uses linear interpolation for remaining gaps. This was derived from our mentor's previous work on the project.
9. Random Forest: Treats the missing price as a target variable, predicting it using features like State, Month, DayOfWeek, and historical lags.
10. SVD Matrix Factorization: Treats the dataset as a Date-by-Mandi matrix and applies Truncated SVD to correlates market behaviors across regions.

Benchmarking Methodology & Analysis

We benchmarked these strategies at $\geq 0\%$, $\geq 50\%$, and $\geq 75\%$ data presence thresholds by masking 10% of known values across 5 random seeds to measure accuracy against a hidden ground truth. Crucially, we optimized for RMSE rather than MAE. While MAE treats all errors equally, RMSE heavily penalizes massive outliers. For our downstream GAT-LSTM model, a few catastrophic errors could poison an entire batch, making RMSE the safer target.

Here is how the main contenders performed:

DOW Ratio: This strategy dominated the $\geq 50\%$ threshold and remained highly competitive at $\geq 75\%$. Once we filtered out mandis with massive multi-month gaps, the remaining sparsity was largely structural (e.g., missing weekends), which this targeted.

Rolling Mean: This won the $\geq 75\%$ threshold for crops like Potato and Wheat. When 75% of data is present, simply averaging the tight surrounding window provides an accurate prediction. However, it struggled at the 50% mark because it could not accurately reflect multi-day weekend dips.

The Spline Pipeline: While it won first place against all mandis (0% threshold) because of its mathematical ability to bridge 6-month gaps, its performance tanked at higher densities. Forcing a smooth curve through dense, volatile data over-smoothed reality and overshot local peaks, causing high RMSE.

The Capped Forward Fill: This method frequently scored the best MAE, meaning it was accurate on typical days, but it consistently lost in RMSE. Carrying stale prices forward created severe outliers that would have derailed our deep learning architecture.

Based on this analysis, we integrated the DOW ratio into our final pipeline for mandis with at least 50% data availability per year. With this clean dataset, we successfully trained a one-day prediction model using LightGBM to establish a baseline for all four crops. We also held a meeting with our mentor to discuss these pipeline updates and model results.

Next Steps

Over the next few weeks, our plan is to:

- Check a broader variety of data availability thresholds, specifically investigating the 50-75% range.

- Look into older historical data, specifically checking for 2022 records.
- Focus on model improvement, developing the architecture for higher multi-day prediction horizons (1-15 days).
- Group the mandis into clusters based on their production volume and test if this clustering affects the model's overall efficacy.